Final Submission

Duolingo APM/SPM Application

Market Overview

The global language-learning market was valued at approximately $28 billion in 2023 and is projected to exceed $90 billion by 2030 (Yahoo Finance). Rapid advances in AI tutoring, voice recognition, and adaptive learning are transforming how learners practice languages, emphasizing personalization and natural conversation. Today's learners increasingly prioritize spoken fluency and real-world communication, yet many apps fail to deliver authentic dialogue or detailed feedback. Common frustrations include repetitive modules, limited feedback specificity, and content disconnected from real-life contexts. These gaps present an opportunity for Duolingo to lead the next wave of interactive, human-centered language learning.

Competitor Analysis

Babbel

- Focuses on structured lessons with cultural and conversational context.
- Caters mainly to adult professionals seeking career or travel proficiency.
- Offers paid-only access, lacking Duolingo's accessibility.
- *Strength:* realistic conversations.
- *Weakness:* limited gamification and low community interaction.

HelloTalk

- Enables one-on-one peer exchanges with native speakers.
- Lacks structured lessons or guided progression.
- User experience varies depending on partner reliability and intent.
- *Strength:* authentic immersion.
- *Weakness:* inconsistent quality and less appealing to beginners.

User Personas

Emma (36)
A nurse learning Spanish to better communicate with her patients through genuine conversation, not scripted phrases. She uses Duolingo but struggles to achieve spoken fluency and understand cultural idioms.

- *Pain Points:* Generic feedback, limited real-life dialogue, lack of cultural nuance.
- *Attributes:* Committed learner, professional motivation, willing to pay for value-added features.

Nate (19)
A college student preparing for a semester abroad in China. He wants a low-stress, engaging way to practice Mandarin pronunciation and tones.

- *Pain Points:* Anxiety about live conversation, repetitive drills, little tone-specific feedback.
- *Attributes:* Free-tier user, values gamification, prefers short, daily learning sessions.

Proposed Features

1. Language Match (Swipe-Based Peer Learning)
Pairs learners with native speakers or peers for short, structured conversation exchanges using a "swipe-to-match" interface that filters by fluency level, goals, and shared interests. This will build confidence and conversational fluency through real, two-way communication while maintaining Duolingo's intuitive and gamified style.

2. Peer Feedback & Rewards System
After each session, users exchange ratings, earn badges, and climb language-specific leaderboards based on helpfulness and consistency. This fosters a safe, motivating community that rewards collaboration and engagement.

3. Open Opportunities Board
A shared community space where users can post or join language events, cultural exchanges, or speaking challenges. This feature extends Duolingo beyond lessons, reinforcing its value as a social and immersive learning hub.

Key Metrics for Success

- *Peer Conversation Rate:* Percent of active users engaging in peer sessions daily will measure adoption and community participation.
- *Retention Rate:* Percent of users returning after 30 days will gauge sustained engagement and satisfaction.
- *Average Session Length:* Time spent in conversational activities will reflect depth of learning and user enjoyment.

Figma

Link Here!

Drafts and workflow!

HARVARD PRODUCT LAB

Pre-Case Team Final Project
Fall 2025

APM/SPM Assignment: New Product Development Case

Workflow paragraph (placed here for clarity)

- I began by referencing online intro courses and noticing general patterns regarding PRD structure, and proceeded to make a bulleted list of important topics to touch upon. (The original project outline is located at the very end.)
- I then did some research online and created a rough draft of my findings and ideas in **red text**.
- Later on, I used ChatGPT for editing grammar mistakes and flow as well as noting any relevant information that's missing in **blue**. (Sometimes I put the text through Chat twice, and that text is in **purple**)
- I decided to take a stab at Figma for the first time, so I did request AI help on the sidebar for navigating the workspace with my ideas.

1. **Market Overview**
   Write 1 short paragraph (3–5 sentences) summarizing the state of the language-learning market. Address key trends, challenges users face in achieving fluency, and competitive opportunities.
   - ***Rough draft:**
   - The language learning market is currently valued at $28billion USD in 2023, and expected to surpass 90 billion in 2030, according to Yahoo finance.
   - AI tutors and speech recognition have been rising, hot features
   - Learners are prioritizing spoken fluency and natural conversational skills
   - Users face challenges with little difficulty in apps (limited answer choices and suggested responses after merely few seconds), limited feedback (usually just "correct" or "incorrect" responses), or feedback that isn't specific to user mistakes, poor sentence flow, limited course material that does not reflect the real world, and modules that become repetitive rather than new and engaging (reddit threads)

***Chat assisted content:**

The global language-learning market was valued at approximately $20 billion in 2023 and is projected to exceed $90 billion by 2030 (Yahoo Finance). Rapid advancements in AI tutoring and speech recognition are reshaping the learning experience, enabling more personalized and conversational practice. Learners today prioritize speaking fluency and real-world communication, seeking tools that mimic natural

conversation. However, common challenges persist with limited feedback specificity, overly simplistic exercises, and stagnant or repetitive content that fails to reflect authentic cultural contexts. These gaps present opportunities for more interactive, adaptive, and human-centered language-learning solutions.

Chat round two:

The global language-learning market was valued at approximately **$20 billion in 2023** and is projected to exceed **$90 billion by 2030** (Yahoo Finance). Rapid advances in **AI tutoring, voice recognition, and adaptive learning** are transforming how learners practice languages, emphasizing personalization and natural conversation. Today's learners increasingly prioritize **spoken fluency** and real-world communication, yet many apps fail to deliver authentic dialogue or detailed feedback. Common frustrations include **repetitive modules**, **limited feedback specificity**, and **content disconnected from real-life contexts**. These gaps present an opportunity for Duolingo to lead the next wave of **interactive, human-centered language learning**.

2. **Competitor Analysis**
   Select 2 major competitors in the language-learning space and provide 3–5 bullet points for each that compare and contrast with Duolingo. Focus on features, target audiences, and unique strengths or weaknesses.

*****Rough draft:**

Babbel

- Emphasizes natural language early on unlike Duolingo
- Offers more structured lessons to teach foundationals unlike Duolingo's game based structure
- Provides context for words and phrases rather than attempting 1:1 translation like Duolingo
- Paid-only features, not free at all unlike Duolingo
- Offers a smaller selection of languages (no Japanese, Korean, Chinese, etc), unlike Duolingo

HelloTalk

- Emphasizes one on one peer conversation very unlike Duolingo
- Does not provide lessons/teaching material unlike Duolingo
- User experience depends on partner (whether they are committed to learning a language or have other motives while using the app) unlike Duolingo's individual based learning

*****Chat assisted content:**

Babbel

- Focuses on **structured lessons** with cultural and conversational context.
- Caters mainly to **adult professionals** seeking career or travel proficiency.
- Offers **paid-only access**, lacking Duolingo's accessibility.
- Prioritizes practical speaking skills but has **limited gamification** and community interaction.
- Provides fewer language options (no Japanese, Korean, or Chinese).

HelloTalk

- Enables **one-on-one peer exchanges** with native speakers through chat and voice.
- Provides **authentic conversation** but no structured teaching materials.
- **User experience varies** based on partner reliability or intent.
- Strong in immersion but **lacks consistency and quality control**.
- Less appealing to casual or beginner learners compared to Duolingo's guided approach.

3. **User Personas**
   Define **two user personas** that represent the typical users of this new feature or product. Each persona should include:
   - A brief description of their background and motivation for learning a language.
   - Their main pain points with current solutions, including Duolingo.
   - A brief summary of key attributes (e.g., learning goals, free vs. paid user, time commitment).

*****Rough draft:**

Emma

- 36, nurse who wants to learn Spanish to better communicate with patients
  - Learning spanish on duo
  - Wants to achieve spoken fluency to help reassure and guide patients through their concerns and make them feel at ease
  - Is concerned about not being able to communicate efficiently and understand the culture/cultural idioms/expressions
  - Working and willing to pay

Nate

- 19, preparing for a study abroad in China
  - Learning Chinese on duo
  - Is highly stressed and broke during the semester so wants low stress, free, fun methods
  - Is concerned about not being understood by native speakers and struggling with tones

*****Chat assisted content:**

Emma (36)

- A nurse learning Spanish to communicate more effectively with her patients. She uses Duolingo but struggles to reach spoken fluency and understand cultural idioms and expressions. Emma wants to make patients feel comfortable through authentic conversation
  - Pain points: Limited real-life dialogue, generic feedback, lack of cultural nuance.
  - Attributes: Committed learner, professional motivation, likely to pay for value-added features.

Nate (19)

- A college student preparing for a study abroad program in China. He seeks a low-stress, engaging way to practice Mandarin amid a busy schedule. Nate worries about pronunciation and tones when speaking with native speakers.
    - Pain points: Anxiety about real conversation, limited tone feedback, repetitive exercises.
    - Attributes: Free-tier user, values fun, gamified experiences, short daily commitment.

Chat round two:

Emma (36)
A nurse learning Spanish to better communicate with her patients. She uses Duolingo but struggles to achieve spoken fluency and understand cultural idioms. Emma wants to comfort patients through genuine conversation, not scripted phrases.

- **Pain Points:** Generic feedback, limited real-life dialogue, lack of cultural nuance.
- **Attributes:** Committed learner, professional motivation, willing to pay for value-added features.

Nate (19)
A college student preparing for a semester abroad in China. He wants a low-stress, engaging way to practice Mandarin pronunciation and tones.

- **Pain Points:** Anxiety about live conversation, repetitive drills, little tone-specific feedback.
- **Attributes:** Free-tier user, values gamification, prefers short, daily learning sessions.

4. **Features**
   Propose **2–3 new features** or improvements that address the pain points identified in the user personas. For each feature, include:
    - A 1-2 sentence description of what it is and how it works
    - A 1-2 sentence justification for how it will improve Duolingo's value proposition or user outcomes

*****Rough draft:**

Some new ideas:

- Swipe based peer learner (tinder for languages)
    - Can chat with native speakers of the same language that you are learning and vice versa

    - Maybe a paid feature where native speakers can simply get paid if they don't receive the service back
-
- Peer feedback system
    - Leaderboard/badge rewards system
        - Peers can give reviews on their sessions and partners
    - Open opportunities board
        - Peers can post opportunities for language calls, culture events,etc

<u>*Chat assisted content:</u>**

Duolingo currently offers gamified lessons in the form of XP points, streaks, leaderboards, an Duolingo Max AI tutor that uses GPT-4 for conversation practice & feedback, Adventures / Quests short story-based mini games that practice comprehension, Podcasts & stories — immersion-style content, and adding New "Speaking & Listening" modules for real-world listening & pronunciation practice.

1. Swipe-Based Peer Learning (Language Match)
A feature that pairs users with native speakers for casual language exchanges, using a familiar "swipe" interface similar to Tinder except that instead of images, it's like a "portfolio" of languages, fluency, and interests/conversation interests. Users can chat, call, or schedule practice sessions.
→ This feature increases **engagement and conversational confidence** by fostering real, two-way communication and broadening users' exposure to authentic speech patterns.

2. Peer Feedback and Rewards System
After each exchange, users can rate and review partners, earning **badges or leaderboard points** for constructive participation.
→ Builds a **motivating community ecosystem** and ensures higher interaction quality while maintaining Duolingo's gamified appeal.

3. Open Opportunities Board
A shared space where learners post or discover **language events, video calls, or cultural exchange opportunities**.
→ Encourages consistent, real-world practice and deepens cultural learning, reinforcing long-term retention and motivation.

<u>Chat round two:</u>

1. Language Match (Swipe-Based Peer Learning)
A feature that pairs learners with native speakers or peers for short, structured conversation exchanges using a familiar "swipe" interface that matches by interests, goals, and fluency level and Duolingo stats.
→ Builds **confidence and conversational fluency** through real, two-way communication while maintaining Duolingo's intuitive design and playful tone.

2. Peer Feedback & Rewards System
After each conversation, users can leave feedback, earn badges, and climb language-specific leaderboards based on participation and helpfulness.
→ Encourages a **motivating and safe community** that rewards positive interaction and accountability.

3. Open Opportunities Board
A shared space for users to post or join **language exchange events**, virtual meetups, or cultural calls.
→ Extends learning beyond the app, reinforcing Duolingo's role as a **gateway to real-world language immersion**.

5. **Key Metrics for Success**
   Identify 3 **key performance indicators (KPIs)** to measure the impact of your features, and give a 1-sentence justification for each.

Peer conversation rates: how many users engaged with other users to have a conversation per day, shows interest in features

Retention rates: how many users stayed on the app and continued using the newly implemented features, shows increased engagement and impact

Session lengths: how much time users spent learning with the newly implemented features, longer, more frequent sessions shows satisfaction

*****Chat assisted content:**

Peer Conversation Rate: Percentage of users engaging in peer sessions daily — measures adoption and community activity.

Retention Rate: Percentage of users continuing to use new features after 30 days — tracks sustained engagement and satisfaction.

Average Session Length: Time spent in peer or conversational activities — indicates depth of learning and enjoyment.

Original Project Outline:

Market Overview

- Clarify Problem, Audience, and Value

Competitor Analysis

- Identify Goals and Key Results

User Personas

- Name
- Goals
- Frustrations
- Quote

Research and Insights

- KPIs
- Methods
- Findings
- Themes

Product Ideas and Solutions

- “What does your app do better”
- Feature
- Problem Solved
- Why it matters
- Must Have
- Should Have
- Could Have
- Won’t Yet Have

Ideas

- Swipe based peer learner (tinder for languages)
- Open opportunities board
- Peer feedback system
    - Leaderboard/badge rewards system